

Trigonometric functions and equations - Set 1 - Solutions

Specification MM4.3-MM4.6.

Answer key

Question	Answer	Topic
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Worked solutions

Question 1 TRG-01 · Trigonometric functions and equations · spec MM4.6

How many solutions does $\sin 2x = \frac{1}{2}$ have for $0 \leq x < 2\pi$?

Answer: C

$2x$ runs over $[0, 4\pi)$, which contains two full periods of $\sin$, each giving two solutions: 4 in total.

Full method - how and why

Counting solutions of $\sin(kx) = c$. Transform the interval for the inside expression and count there.

- 1 Let $u = 2x$. As x runs over $[0, 2\pi)$, u runs over $[0, 4\pi)$.
why: Multiplying the variable by 2 doubles the range of the inside expression - this is the whole trick.
- 2 $[0, 4\pi)$ contains exactly two full periods of $\sin$.
why: $\sin$ has period 2π , so a 4π -long interval holds two complete waves.
- 3 In each period, $\sin u = \frac{1}{2}$ has two solutions (one on the way up, one on the way down).
why: A horizontal line strictly between -1 and 1 cuts each wave twice.
- 4 Total: $2 \times 2 = 4$ solutions.
why: Each u gives exactly one x via $x = u/2$, so no solutions are lost or duplicated.

Where it goes wrong

- Answering 2 by counting solutions for x as if the 2 were not there.
- Answering 8 by doubling twice.

Takeaway For $\sin(kx) = c$ on an interval of length L , the inside runs over length kL - count solutions there.

Question 2 TRG-03 · Trigonometric functions and equations · spec MM4.3

Find the exact value of $\sin 60^\circ \cos 30^\circ + \cos 60^\circ \sin 30^\circ$.

Answer: D

$$\frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{4} + \frac{1}{4} = 1.$$

Full method - how and why

Pure recall of exact values. (It is also $\sin(60^\circ + 30^\circ) = \sin 90^\circ$, but the addition formula is off-specification.)

- 1 Write down the exact values: $\sin 60^\circ = \frac{\sqrt{3}}{2}$, $\cos 30^\circ = \frac{\sqrt{3}}{2}$, $\cos 60^\circ = \frac{1}{2}$, $\sin 30^\circ = \frac{1}{2}$.
why: These come from the half-equilateral triangle with sides 1, $\sqrt{3}$, 2 - worth being able to redraw in seconds.
- 2 Substitute: $\frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{4} + \frac{1}{4}$.
why: $(\sqrt{3})^2 = 3$, so the first product is $\frac{3}{4}$.
- 3 Add: 1.
why: A clean answer - the sign that the exact values were right.

Where it goes wrong

- Using $\sin 60^\circ = \frac{1}{2}$ (that is $\sin 30^\circ$), giving $\frac{\sqrt{3}}{2}$.
- Writing $\left(\frac{\sqrt{3}}{2}\right)^2 = \frac{\sqrt{3}}{4}$.

Takeaway $\sin$ and $\cos$ of 30° and 60° swap over; square a surd fraction by squaring top and bottom.

Question 3 TRG-04 · Trigonometric functions and equations · spec MM4.6

Find the sum of all solutions of $\tan x = \sqrt{3}$ in the interval $0 \leq x \leq 2\pi$.

Answer: D

$$x = \frac{\pi}{3} \text{ and } x = \frac{\pi}{3} + \pi = \frac{4\pi}{3}. \text{ Sum } \frac{5\pi}{3}.$$

Full method - how and why

$\tan$ has period π , so solutions come one per period. Find the first, add π , then check the interval.

- 1 The principal solution: $\tan x = \sqrt{3}$ at $x = \frac{\pi}{3}$.
why: From the exact values: $\tan 60^\circ = \sqrt{3}$.
- 2 $\tan$ repeats every π , so the next solution is $\frac{\pi}{3} + \pi = \frac{4\pi}{3}$.
why: Unlike $\sin$ and $\cos$, $\tan$ has period π , not 2π .
- 3 Check the interval $[0, 2\pi]$: $\frac{\pi}{3}$ and $\frac{4\pi}{3}$ are in it; the next, $\frac{7\pi}{3}$, is not.
why: $\frac{7\pi}{3} > 2\pi$, so stop there.
- 4 Sum: $\frac{\pi}{3} + \frac{4\pi}{3} = \frac{5\pi}{3}$.
why: Common denominator 3.

Where it goes wrong

- Using period 2π and finding only $\frac{\pi}{3}$.
- Including $\frac{7\pi}{3}$ and answering $\frac{7\pi}{3}$ or more - which is offered.

Takeaway $\tan$ repeats every π : solutions are $x_0 + k\pi$, so there are usually two in a 2π interval.

Question 4 TRG-06 · Trigonometric functions and equations · spec MM4.5

Given that $\sin \theta = \frac{3}{5}$ and θ is obtuse, find $\tan \theta$.

Answer: A

$$\cos \theta = -\frac{4}{5} \text{ (negative in the second quadrant), so } \tan \theta = \frac{3/5}{-4/5} = -\frac{3}{4}.$$

Full method - how and why

From one ratio to another, with a quadrant condition fixing the signs. Use a right-angled triangle plus CAST.

- 1 From $\sin \theta = \frac{3}{5}$, the reference triangle has opposite 3 and hypotenuse 5, so the third side is 4 (a 3-4-5 triangle).
why: Pythagoras gives the missing side; the 3-4-5 triple appears constantly.
- 2 θ is obtuse, so it lies in the second quadrant where $\sin$ is positive but $\cos$ and $\tan$ are negative.
why: CAST: only sine is positive in the second quadrant.
- 3 So $\cos \theta = -\frac{4}{5}$ and $\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{3/5}{-4/5} = -\frac{3}{4}$.
why: The fives cancel; the sign comes from the quadrant.

Where it goes wrong

- Ignoring 'obtuse' and answering $\frac{3}{4}$.
- Inverting the ratio and answering $-\frac{4}{3}$.

Takeaway Build the reference triangle from the given ratio, then let the quadrant decide the signs.

Question 5 TRG-07 · Trigonometric functions and equations · spec MM4.6

Solve $\sin^2 x = \frac{3}{4}$ for $0 \leq x < 2\pi$. How many solutions are there?

Answer: E

$$\sin x = \pm \frac{\sqrt{3}}{2}: x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}. \text{ Four solutions.}$$

Full method - how and why

A squared trig equation. Taking the square root produces **two** values, each with its own solutions.

1 Take square roots: $\sin x = \pm \frac{\sqrt{3}}{2}$.

why: $X^2 = k$ has two roots - forgetting the negative one halves the answer.

2 $\sin x = \frac{\sqrt{3}}{2}$ gives $x = \frac{\pi}{3}$ and $\frac{2\pi}{3}$.

why: Second-quadrant partner: $\pi - \frac{\pi}{3}$.

3 $\sin x = -\frac{\sqrt{3}}{2}$ gives $x = \frac{4\pi}{3}$ and $\frac{5\pi}{3}$.

why: Third and fourth quadrants, by symmetry below the axis.

4 Total: 4 solutions in $[0, 2\pi)$.

why: Count them; symmetry says there should be an even number here.

Where it goes wrong

- Taking only the positive root and answering 2.
- Reading $\sin^2 x$ as $\sin(x^2)$.

Takeaway $\sin^2 x = c$ gives $\sin x = \pm \sqrt{c}$, and each sign contributes its own solutions.

Question 6 TRG-08 · Trigonometric functions and equations · spec MM4.6

Find the smallest positive solution of $\cos\left(x - \frac{\pi}{6}\right) = -\frac{1}{2}$.

Answer: B

$$x - \frac{\pi}{6} = \frac{2\pi}{3} \text{ gives } x = \frac{5\pi}{6} \text{ (the other solution } x - \frac{\pi}{6} = \frac{4\pi}{3} \text{ gives } \frac{3\pi}{2}, \text{ larger).}$$

Full method - how and why

An equation with a shifted argument. Solve for the bracket first, then unshift - and be careful about which solution is smallest.

1 Let $u = x - \frac{\pi}{6}$ and solve $\cos u = -\frac{1}{2}$.

why: Substituting keeps the shift out of the way until the end.

2 $\cos u = -\frac{1}{2}$ at $u = \frac{2\pi}{3}$ and $u = \frac{4\pi}{3}$ (second and third quadrants).

why: $\cos$ is negative in quadrants 2 and 3; the reference angle is $\frac{\pi}{3}$.

3 Convert back: $x = u + \frac{\pi}{6}$, giving $x = \frac{2\pi}{3} + \frac{\pi}{6} = \frac{5\pi}{6}$ and $x = \frac{4\pi}{3} + \frac{\pi}{6} = \frac{3\pi}{2}$.

why: Common denominator 6: $\frac{4\pi}{6} + \frac{\pi}{6} = \frac{5\pi}{6}$.

4 The smallest positive one is $\frac{5\pi}{6}$.

why: Compare the candidates; $\frac{5\pi}{6} < \frac{3\pi}{2}$.

Where it goes wrong

- Subtracting the shift instead of adding when converting back.
- Using the reference angle $\frac{\pi}{3}$ directly, which gives $\frac{\pi}{2}$ - an offered option.

Takeaway Solve for the whole bracket, then undo the shift; adjust the interval for the bracket if the question restricts x .

Question 7 TRG-09 · Trigonometric functions and equations · spec MM4.5

Simplify $\frac{1 - \cos^2 \theta}{\cos \theta \tan \theta}$ for values of θ where the expression is defined.

Answer: B

$\cos \theta \tan \theta = \sin \theta$, and $1 - \cos^2 \theta = \sin^2 \theta$. So the expression is $\sin \theta$.

Full method - how and why

Simplification using the two permitted identities. Work on the denominator and numerator separately.

- 1 Denominator: $\cos \theta \tan \theta = \cos \theta \times \frac{\sin \theta}{\cos \theta} = \sin \theta$.
why: $\tan \theta = \frac{\sin \theta}{\cos \theta}$ and the $\cos \theta$ cancels (legitimate where the expression is defined).
- 2 Numerator: $1 - \cos^2 \theta = \sin^2 \theta$.
why: Rearranged Pythagorean identity.
- 3 Divide: $\frac{\sin^2 \theta}{\sin \theta} = \sin \theta$.
why: One power of $\sin \theta$ cancels.

Where it goes wrong

- Cancelling to $\sin^2 \theta$ by forgetting to simplify the denominator.
- Writing $1 - \cos^2 \theta = \cos^2 \theta - 1$ (sign flip), giving $-\sin \theta$.

Takeaway Convert $\tan$ into $\frac{\sin}{\cos}$ and use $1 - \cos^2 = \sin^2$; then cancel.

Question 8 TRG-11 · Trigonometric functions and equations · spec MM4.4

The function $f(x) = 3\sin(2x) - 1$ is defined for all real x . Find the difference between its maximum and minimum values.

Answer: E

Range is $[-4, 2]$; the difference is 6.

Full method - how and why

Range of $a\sin(bx) + c$. Only a and c matter - the b affects the period, not the values taken.

- 1 $\sin(2x)$ takes every value in $[-1, 1]$.
why: Compressing horizontally changes how fast the wave oscillates, not how high it goes.
- 2 Multiply by 3: the range becomes $[-3, 3]$.
why: Amplitude $|a| = 3$.
- 3 Subtract 1: the range becomes $[-4, 2]$.
why: The whole wave shifts down by 1.
- 4 Difference between maximum and minimum: $2 - (-4) = 6$.
why: Equivalently $2 \times \text{amplitude} = 2 \times 3 = 6$ - the vertical shift cancels out of a difference.

Where it goes wrong

- Answering 3 (the amplitude) instead of the full range $2a$.
- Letting the -1 affect the difference, giving 7 or 5.

Takeaway Max minus min for $a\sin(\cdot) + c$ is always $2|a|$; vertical shifts cancel.

Question 9 TRG-12 · Trigonometric functions and equations · spec MM4.6

Find the number of solutions of $\tan^2 x = 3$ for $0 \leq x < 2\pi$.

Answer: B

$\tan x = \pm\sqrt{3}$: $x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$. Four solutions.

Full method - how and why

A squared tangent equation. Two values of $\tan x$, each giving solutions once per π .

1 Take square roots: $\tan x = \pm\sqrt{3}$.

why: Both signs are needed.

2 $\tan x = \sqrt{3}$ gives $x = \frac{\pi}{3}$ and $\frac{\pi}{3} + \pi = \frac{4\pi}{3}$.

why: Period π for $\tan$.

3 $\tan x = -\sqrt{3}$ gives $x = \frac{2\pi}{3}$ and $\frac{2\pi}{3} + \pi = \frac{5\pi}{3}$.

why: $\tan$ is negative in quadrants 2 and 4; the reference angle is still $\frac{\pi}{3}$.

4 Four solutions in $[0, 2\pi)$.

why: Two values $\times$ two periods.

Where it goes wrong

- Taking only the positive root and answering 2.
- Using period 2π for $\tan$.

Takeaway $\tan^2 x = k$ gives $\tan x = \pm\sqrt{k}$, and each has one solution per interval of length π .

Question 10 TRG-13 · Trigonometric functions and equations · spec MM4.5

For all θ where the expressions are defined, $(\sin \theta + \cos \theta)^2 + (\sin \theta - \cos \theta)^2$ equals

Answer: D

Expanding: $(1 + 2\sin \theta \cos \theta) + (1 - 2\sin \theta \cos \theta) = 2$.

Full method - how and why

Expanding two brackets whose cross terms cancel. The Pythagorean identity finishes it.

1 Expand the first: $(\sin \theta + \cos \theta)^2 = \sin^2 \theta + 2\sin \theta \cos \theta + \cos^2 \theta$.

why: Standard $(a + b)^2$ expansion.

2 Expand the second: $(\sin \theta - \cos \theta)^2 = \sin^2 \theta - 2\sin \theta \cos \theta + \cos^2 \theta$.

why: Only the middle sign changes.

3 Add: the $\pm 2\sin \theta \cos \theta$ terms cancel, leaving $2(\sin^2 \theta + \cos^2 \theta)$.

why: This cancellation is the point of the question.

4 Apply the identity: $2 \times 1 = 2$.

why: $\sin^2 + \cos^2 = 1$ for every θ , so the answer is a constant.

Where it goes wrong

- Keeping a cross term and choosing $2 + 2\sin \theta \cos \theta$ - an offered option.
- Answering 1 by forgetting the factor 2.

Takeaway $(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$; with $\sin$ and $\cos$ that is always 2.

Question 11 TRG-16 · Trigonometric functions and equations · spec MM4.6

Given that $\sin x + \cos x = 0$ and $0 \leq x < 2\pi$, find the sum of all possible values of x .

Answer: B

$$\tan x = -1: x = \frac{3\pi}{4} \text{ or } \frac{7\pi}{4}. \text{ Sum} = \frac{10\pi}{4} = \frac{5\pi}{2}.$$

Full method - how and why

Convert a sum of sin and cos into a single tan equation by dividing.

- 1 Rearrange: $\sin x = -\cos x$.
why: Move one term across.
- 2 Divide by $\cos x$: $\tan x = -1$. (Check $\cos x = 0$ separately: then $\sin x = 0$ too, impossible.)
why: Dividing by $\cos x$ is safe only after ruling out $\cos x = 0$ - and here that case gives a contradiction.
- 3 Solve on $[0, 2\pi)$: $\tan x = -1$ at $x = \frac{3\pi}{4}$ and $x = \frac{3\pi}{4} + \pi = \frac{7\pi}{4}$.
why: Reference angle $\frac{\pi}{4}$, negative tangent in quadrants 2 and 4.
- 4 Sum: $\frac{3\pi}{4} + \frac{7\pi}{4} = \frac{10\pi}{4} = \frac{5\pi}{2}$.
why: Add over the common denominator and simplify.

Where it goes wrong

- Finding only one solution and answering $\frac{3\pi}{4}$ - an offered option.
- Forgetting to justify dividing by $\cos x$ (here it is fine, but the habit matters).

Takeaway $a \sin x + b \cos x = 0$ becomes $\tan x = -\frac{b}{a}$; check the $\cos x = 0$ case before dividing.

Question 12 TRG-18 · Trigonometric functions and equations · spec MM4.3

Find the value of $\tan 30^\circ \tan 60^\circ + \sin^2 45^\circ$.

Answer: C

$$\tan 30^\circ \tan 60^\circ = \frac{1}{\sqrt{3}} \cdot \sqrt{3} = 1, \sin^2 45^\circ = \frac{1}{2}. \text{ Total } \frac{3}{2}.$$

Full method - how and why

Exact-value recall plus one squared value. The first product is designed to collapse to 1.

- 1 $\tan 30^\circ = \frac{1}{\sqrt{3}}$ and $\tan 60^\circ = \sqrt{3}$, so their product is 1.
why: 30° and 60° are complementary, and $\tan \theta \tan(90^\circ - \theta) = 1$ always.
- 2 $\sin 45^\circ = \frac{\sqrt{2}}{2}$, so $\sin^2 45^\circ = \frac{2}{4} = \frac{1}{2}$.
why: Square top and bottom: $(\sqrt{2})^2 = 2$ and $2^2 = 4$.
- 3 Add: $1 + \frac{1}{2} = \frac{3}{2}$.
why: Straightforward once both pieces are exact.

Where it goes wrong

- Writing $\sin^2 45^\circ = \frac{\sqrt{2}}{2}$ (forgetting to square), giving $1 + \frac{\sqrt{2}}{2}$.
- Using $\tan 30^\circ = \sqrt{3}$, which makes the product 3.

Takeaway $\tan \theta \tan(90^\circ - \theta) = 1$, and $\sin^2 45^\circ = \cos^2 45^\circ = \frac{1}{2}$.